Literature Review: Virtual Try-On Systems Using Deep Learning

March 10, 2026

Contents

1 Introduction

Virtual Try-On (VTON) systems aim to generate realistic images of a person wearing a new garment using deep learning and computer vision techniques. With the rapid growth of e-commerce in the fashion industry, VTON technology has become increasingly important for improving customer experience and reducing product return rates. The primary goal of these systems is to realistically overlay a clothing item from a product image onto a person while preserving body pose, identity, and garment details.

However, achieving realistic virtual try-on remains challenging due to several factors including garment deformation, pose variation, occlusions, texture preservation, and alignment between clothing and the human body. Over the past few years, researchers have proposed multiple approaches including GAN-based methods, flow-based warping models, and diffusion-based generative models.

This literature review analyzes seven major research papers that contributed to the development of virtual try-on systems.

2 Review of Existing Work

2.1 VITON – An Image-Based Virtual Try-On Network

Authors: Xintong Han et al.
Year: 2018
Conference: CVPR

Paper Links

- `https://arxiv.org/abs/1711.08447`
- `https://openaccess.thecvf.com/content_cvpr_2018/papers/Han_VITON_An_Image-Based_ CVPR_2018_paper.pdf`

Problem Statement

The paper addresses the problem of generating realistic images of a person wearing a target clothing item using only RGB images without relying on 3D body modeling techniques.

Methodology

The proposed framework follows a coarse-to-fine strategy consisting of:

1. Clothing-agnostic person representation
2. Encoder–decoder generator network
3. Garment warping module
4. Refinement network

Dataset Used

- VITON dataset

Results

The model generates realistic images while maintaining the person's pose and clothing appearance.

Limitations

- Low resolution outputs
- Poor preservation of fine garment textures
- Alignment issues for complex poses

Contribution

Introduced the first deep learning based virtual try-on system using only image inputs without requiring 3D body models.

2.2 Pix2Surf – Learning to Transfer Texture from Clothing Images to 3D Humans

Authors: Aymen Mir et al.
Year: 2020
Conference: CVPR

Paper Link

- `https://arxiv.org/abs/2003.02050`

Problem Statement

The goal of this work is to automatically transfer clothing textures from garment images onto 3D garment models in real time.

Methodology

The Pix2Surf model learns dense correspondences between clothing images and 3D garment surfaces. The pipeline includes:

1. Non-rigid alignment of garment templates
2. Mapping garment silhouettes to UV coordinates
3. Texture transfer onto 3D human models

Dataset Used

Garment datasets containing aligned 3D templates.

Results

The model enables real-time mapping of garment textures to 3D humans.

Limitations

- Requires predefined garment templates
- Limited ability to handle complex body poses

Contribution

Introduced real-time 3D texture transfer techniques for virtual try-on applications.

2.3 ClothFlow – A Flow-Based Model for Clothed Person Generation

Authors: Yuying Ge et al.
Year: 2019
Conference: ICCV

Paper Link

- `https://arxiv.org/abs/1908.07850`

Problem Statement

Existing GAN-based models struggle to accurately align garments with human body poses.

Methodology

ClothFlow introduces appearance flow estimation for garment warping. The pipeline includes:

1. Human parsing
2. Appearance flow estimation
3. Garment warping
4. Image synthesis

Dataset Used

- VITON dataset

Results

The approach significantly improves garment alignment compared to previous GAN-based methods.

Limitations

- Computationally expensive
- Warping artifacts may still occur

Contribution

Introduced flow-based garment warping for virtual try-on systems.

2.4 VITON-HD – High Resolution Virtual Try-On

Authors: Seunghwan Choi et al.
Year: 2021
Conference: CVPR

Paper Link

- `https://arxiv.org/abs/2103.16874`

Problem Statement

Earlier VTON models generated low-resolution images that were insufficient for real-world applications.

Methodology

The authors introduce:

- Clothing-agnostic person representation
- ALIAS normalization
- ALIAS generator

These components improve texture preservation and reduce misalignment artifacts.

Dataset Used

- VITON-HD dataset

Resolution: 1024×768

Results

The system produces significantly higher resolution and more detailed try-on images.

Limitations

- Computationally heavy training
- Misalignment issues may still occur

Contribution

Introduced the first high-resolution virtual try-on framework.

2.5 GP-VTON – General Purpose Virtual Try-On

Authors: Zhenyu Xie et al.
Year: 2023
Conference: CVPR

Paper Link

- https://arxiv.org/abs/2305.16086

Problem Statement
Existing models struggle with diverse clothing styles and complex human poses.

Methodology
GP-VTON introduces collaborative learning between:

- Local flow estimation
- Global body parsing

Dataset Used

- VITON-HD dataset

Results
Improves garment alignment and overall realism.

Limitations

- Boundary artifacts may appear
- Complex architecture

Contribution
Introduced collaborative flow and parsing learning.

2.6 LaDI-VTON – Latent Diffusion Virtual Try-On

Authors: Davide Morelli et al.
Year: 2023
Conference: ACM Multimedia

Paper Link

- https://arxiv.org/abs/2307.05177

Problem Statement
GAN-based models often struggle to produce highly realistic outputs.
Methodology
Uses latent diffusion models conditioned on clothing images and textual prompts.
Dataset Used

- VITON-HD dataset

Results
Produces more realistic images with improved perceptual quality.
Limitations

- Slow inference time
- High GPU requirements

Contribution
Introduced diffusion models for virtual try-on image generation.

2.7 IDM-VTON – Improving Diffusion Models for Authentic Virtual Try-On

Authors: Yisol Choi et al.
Year: 2024
Conference: ECCV
Paper Link

- `https://arxiv.org/abs/2403.05139`

Problem Statement
Diffusion-based methods often fail to preserve garment identity and fine textures.
Methodology
Introduces two conditioning modules:

1. Image Prompt Adapter
2. GarmentNet

Dataset Used

- VITON-HD
- DressCode dataset

Results

Improves garment fidelity and realism.

Limitations

- High GPU usage
- Slow inference

Contribution

Introduced dual-conditioning diffusion architecture for better garment preservation.

3 Comparative Analysis

Paper	Year	Method	Strength	Weakness
VITON	2018	GAN	First VTON framework	Low resolution
Pix2Surf	2020	3D Texture Transfer	Real-time 3D try-on	Requires templates
ClothFlow	2019	Flow Based	Better alignment	Expensive computation
VITON-HD	2021	GAN	High resolution	Heavy model
GP-VTON	2023	Flow + Parsing	Better warping	Boundary artifacts
LaDI-VTON	2023	Diffusion	Realistic results	Slow inference
IDM-VTON	2024	Diffusion	Better garment fidelity	High GPU cost

4 Research Gap

Despite significant improvements, several challenges remain:

- Accurate garment alignment with human pose
- Preservation of fine texture details
- High computational cost of diffusion models
- Slow inference time
- Limited generalization to real-world scenarios

5 Conclusion

This review analyzed seven major research papers on virtual try-on systems. Early works focused on GAN-based architectures, while later models introduced flow-based warping methods to improve garment alignment. Recently, diffusion-based approaches have become popular due to their ability to produce more realistic images. However, these models often require high computational resources and still struggle with garment fidelity.

Future research should focus on combining efficient generation methods with improved garment alignment techniques to create realistic and scalable virtual try-on systems.